Task: GIE - Stock Market

Lab 7. Available memory: 512 MB. **Date: 14.12.2025**

Description

Professor Bajtek became interested in analyzing stock market prices in Bitotia. He managed to collect historical data regarding the stock index rate $(a_1, ..., a_n)$ and now needs to perform an analysis.

Bajtek must answer a series of questions regarding the stock index rate. Each question involves determining for a time range $[s, ..., e]$ and a value range $[l, ..., h]$:

1. When is the first time within the time range $[s, ..., e]$ that the index rate falls within the range $[l, ..., h]$ (or state that this never happens):

   $$Q(s, e, l, h) = \min\{i : s \leq i \leq e \text{ and } l \leq a_i \leq h\}$$

   or -1 if no indices satisfy the conditions.

2. How many times within the time range $[s, ..., e]$ is the index rate within the range $[l, ..., h]$:

   $$C(s, e, l, h) = |\{i : s \leq i \leq e \text{ and } l \leq a_i \leq h\}|$$

Input

The first line contains two integers n and m ($1 \leq n \leq 200\,000$, $1 \leq m \leq 300\,000$), representing the length of the history and the number of queries. The second line contains n integers, each in the range $[0, ..., 10^9]$, representing the history of the stock index $a_1, ..., a_n$. The next m lines contain the queries; each line contains four integers s_i, e_i, l_i, h_i ($1 \leq s_i \leq e_i \leq n$, $0 \leq l_i \leq h_i \leq 10^9$).

Output

The program should print m lines. In the i-th line, print two integers separated by a space: the result of the query $Q(s_i, e_i, l_i, h_i)$ and $C(s_i, e_i, l_i, h_i)$

Example

Input:

```
10 4
1 2 10 5 1 2 3 8 9 1000
1 10 2 8
4 9 10 20
6 10 10 2000
1 5 3 6
```

Output:

```
2 5
-1 0
10 1
4 1
```